Laboratory IV
Moment of Inertia and Oscillations

Department of Physics & Astronomy

PHYSICS 113 - 121 - 181

General Physics I and Mechanics

Objective

Measure the moment of inertia of three objects about a specified rotational axis using a photogate timer, and verify the parallel axis theorem. In the second part, investigate Hooke's Law by measuring the oscillation period of a spring-mass system as a function of the attached mass.

Theory

Moment of Inertia and the Parallel Axis Theorem

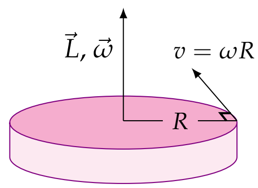

Figure 1: *A rotating solid disk.*

Consider a rigid body rotating about an axis, as in Fig. 1. If the angular velocity is ω, a point in the body at a perpendicular distance r from the axis of rotation will move with linear speed $v = \omega r$. The total angular momentum L of the rotating body points along the axis of rotation and is equal to

$$L = \int_{\text{body}} rv\, dm = \int_{\text{body}} r^2 \omega\, dm = \omega \int_{\text{body}} r^2\, dm = I\omega, \tag{1}$$

where

$$I = \int_{\text{body}} r^2\, dm \tag{2}$$

is called the **moment of inertia** of the body about the axis of rotation. In the SI system of units, I is expressed in units of kg m^2.

If the axis of rotation passes through the center of mass of the object, then I is called I_{CM}. For example, for the solid disk of mass M and radius R shown in Fig. 1,

$$I_{\text{CM}} = \frac{1}{2} MR^2.$$

Table 1 shows I_{CM} for several common mass distributions.

Table 1: *Moments of inertia of several mass distributions.*

Object	**Rotational Axis**	$\mathbf{I_{CM}}$
Thin Ring	Symmetry Axis	MR^2
Thick Ring	Symmetry Axis	$\frac{1}{2}M\left(R_1^2 + R_2^2\right)$
Solid Disk	Symmetry Axis	$\frac{1}{2}MR^2$
Thin Spherical Shell	About a Diameter	$\frac{2}{3}MR^2$
Solid Sphere	About a Diameter	$\frac{2}{5}MR^2$

The **parallel axis theorem**[1] relates I_{CM} to the moment of inertia I about any parallel axis displaced from the center of mass. The theorem states

$$I = I_{\text{CM}} + Md^2, \tag{3}$$

where M is the body's mass and d is the perpendicular distance between the two axes. This implies that I_{CM} is the minimum moment of inertia among all parallel axes.

[1] Also called the Huygens–Steiner theorem, after Christiaan Huygens (1629–1695) and Jakob Steiner (1796–1863).

Analogies between Rotational and Linear Motion

When working with rotational motion for rigid bodies, many of the equations are analogous to those of linear kinematics. For example, angular velocity plays the role of linear velocity, and moment of inertia plays the role of mass. Table 2 summarizes the correspondence.

Table 2: *Analogies between linear and rotational motion.*

Linear Kinematics	**Rotational Kinematics**
Linear Velocity $= v$	Angular Velocity $= \omega$
Mass $= M$	Moment of Inertia $= I$
Linear Momentum $= p = Mv$	Angular Momentum $= L = I\omega$
Kinetic Energy $= K = \frac{1}{2}Mv^2$	Kinetic Energy $= K = \frac{1}{2}I\omega^2$

Determining the Moment of Inertia of an Object

The apparatus in Fig. 2 consists of a rotary table on which you can mount an object in order to measure its moment of inertia. A torsion spring restricts the motion of the table and provides a restoring torque τ. If the table is rotated by an angle θ, then the torque acting on it is

$$\tau = -\kappa\theta, \tag{4}$$

where κ is the torsion spring constant, analogous to the spring constant k for compressed and stretched springs. The value of κ must be measured.

If the table has moment of inertia I_{tab} and an object with moment of inertia I_{obj} is mounted on it, the combined system performs rotary oscillations with angular frequency

$$\omega = \sqrt{\frac{\kappa}{I_{\text{obj}} + I_{\text{tab}}}}, \tag{5}$$

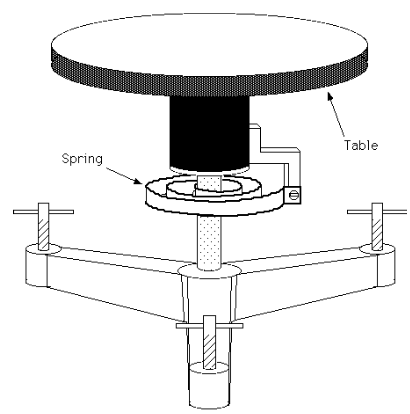

Figure 2: *Rotary table with torsion spring.*

which corresponds to a period of oscillation

$$T = 2\pi\sqrt{\frac{I_{\text{obj}} + I_{\text{tab}}}{\kappa}}. \tag{6}$$

The two unknown parameters are I_{tab} and κ. To determine their values, you will measure the oscillation period of the table alone (T_{tab}) and the period of the table together with an object of known moment of inertia I_{obj} ($T_{\text{tab+obj}}$). From eq. (6), one finds that

$$T_{\text{tab}} = 2\pi\sqrt{\frac{I_{\text{tab}}}{\kappa}}, \tag{7}$$

$$T_{\text{tab+obj}} = 2\pi\sqrt{\frac{I_{\text{obj}} + I_{\text{tab}}}{\kappa}}. \tag{8}$$

To solve for κ, we square and subtract eqs. (7) and (8):

$$(T_{\text{tab+obj}})^2 + (T_{\text{tab}})^2 = 4\pi^2\frac{2I_{\text{tab}} + I_{\text{obj}}}{\kappa}, \tag{9}$$

$$(T_{\text{tab+obj}})^2 - (T_{\text{tab}})^2 = 4\pi^2\frac{I_{\text{obj}}}{\kappa}. \tag{10}$$

Eq. (10) lets us isolate the torsion spring constant κ in terms of known and measured quantities:

$$\boxed{\kappa = 4\pi^2 \frac{I_{\text{obj}}}{(T_{\text{tab+obj}})^2 - (T_{\text{tab}})^2}} \tag{11}$$

To find the unknown I_{tab}, divide eq. (9) by eq. (10):

$$\frac{(T_{\text{tab}})^2 + (T_{\text{tab+obj}})^2}{(T_{\text{tab+obj}})^2 - (T_{\text{tab}})^2} = \frac{2I_{\text{tab}}}{I_{\text{obj}}} + 1, \tag{12}$$

which simplifies to

$$\boxed{I_{\text{tab}} = I_{\text{obj}} \frac{(T_{\text{tab}})^2}{(T_{\text{tab+obj}})^2 - (T_{\text{tab}})^2}} \tag{13}$$

Once κ and I_{tab} are known, the moment of inertia I_x of any other object x can be found from

$$\boxed{I_x = I_{\text{tab}} \left[\left(\frac{T_{\text{tab}+x}}{T_{\text{tab}}} \right)^2 - 1 \right]}, \tag{14}$$

provided we also measure the oscillation period $T_{\text{tab}+x}$ of the table with the new object mounted.

Hooke's Law and Spiral Spring Oscillations

It is often assumed that a long spiral spring obeys Hooke's law when it is not stretched too far. If the spring is hung vertically from a fixed support and a mass is attached to its free end, the mass can be set into simple harmonic motion by stretching and releasing it. The period of oscillation is the time required for the attached mass to return to its initial position. The period T depends on the attached mass M, the spring force constant k, and the spring mass m, and is given by

$$T = 2\pi \sqrt{\frac{M + bm}{k}}. \tag{15}$$

In eq. (15), b is a dimensionless constant called the spring mass coefficient. You will calculate b in this lab and compare it to the theoretical value

$$b_{\text{theory}} = \frac{1}{3}. \tag{16}$$

During the lab, you will measure T for a variety of values of M. To make it easy to estimate k from T and M, square eq. (15):

$$T^2 = \frac{4\pi^2}{k} M + \frac{4\pi^2}{k} bm. \tag{17}$$

A plot of T^2 versus M should therefore yield a straight line of the form

$$y = (\text{slope})x + (y\text{-intercept}). \tag{18}$$

You will plot the data in this manner and determine whether the expected linear relationship holds.

Experiment

Moment of Inertia and Parallel Axis Theorem

Procedure

1. Set up the rotary table and allow it to come to its equilibrium position. Make sure the tabletop is level by placing a level onto it and adjusting the apparatus' feet as needed.

2. Set the photogate timer to pendulum mode and arrange the photogate so the table's trigger (see Fig. 3) passes through the photogate when the table rotates.

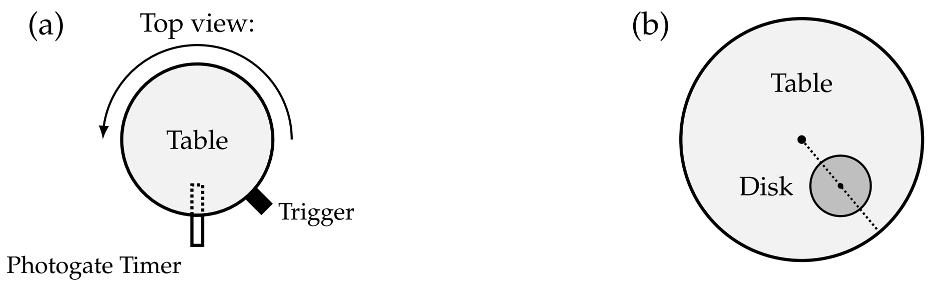

Figure 3: *(a) view of rotary table, photogate timer, and trigger; (b) the table with the disk at an offset from the axis of rotation.*

3. Wind the table either clockwise or counter-clockwise such that *the trigger does not pass the timer while winding*. Stop when the table hits the block and cannot be turned any further.

4. Release the table. Count the number of times the trigger passes through the photogate before the timer stops counting. It should be three passes before the timer stops. The displayed time will be the period of one oscillation.

5. Measure the mass and inner and outer radii of the *brass ring*. Record your values in Data Table 1 in the Postlab Exercises.

6. Measure the period of one oscillation *of the table alone* (T_{tab}) five (5) times. Record your data in Data Table 1 in the Postlab Exercises.

7. After ensuring the brass ring is mounted securely with screws to the tabletop and is centered on the table's axis of rotation, measure the period of oscillation *of the table/ring combination* ($T_{\text{tab+ring}}$) five (5) times. Record the data in Data Table 1.

8. Measure the mass and radius of the solid disk, and record your measurements in Data Table 1.

9. To measure the period of oscillation for the table/disk combination, secure the disk at one of five equally spaced positions along the table radius, a distance d from the table's axis of rotation (see Fig. 3). Start at the center ($d = 0.000\,\text{m}$). The recommended positions are 0.000 m, 0.015 m, 0.030 m, 0.045 m, and 0.060 m, using the spacing holes on the table.

10. For each of the five positions d, measure the period of oscillation five (5) times. Record your data in Data Table 2.

Hooke's Law and Spiral Spring Oscillations

Introduction

According to Hooke's law, the extension x of a stretched spring from its equilibrium position should be proportional to the applied force F,

$$F = -kx, \tag{19}$$

where k is the *force constant* of the spring (also called the *spring constant*). You will test the validity of Hooke's law by measuring x while incrementally increasing the attached mass M. Then, you will measure the period of oscillation for several hanging masses to estimate the spring mass coefficient b needed to account for the spring's own mass m, as described in eqs. (15) and (16).

Procedure

1. Weigh the Slinky Jr.™ and mass holder to determine the mass of the spring/slinky system m. Record it in Data Table 4 in the Postlab Exercises.

2. Mount the spring by clipping one end of it to the flat metal piece mounted to the stand.

3. For each of the attached masses M listed in Data Table 4, calculate the force of gravity $F = Mg$, where $g = 9.8\,\mathrm{m\,s^{-2}}$. Record your calculations in Data Table 4.

4. Let the spring come to equilibrium. If possible, move the ruler/spring vertically so that the ruler's zero is at the bottom of the spring (or the bottom of the mass holder, if you prefer). Record the equilibrium position of the spring's bottom end on the ruler.

5. Measure the change in extension of the spring, x, as you attach different amounts of mass to the end of the spring. The extension is measured relative to the equilibrium length of the unloaded spring. Use masses of 5 g, 10 g, 15 g, 20 g, and 25 g, and attach each mass gently. Record all results in Data Table 4.

6. For each of the five masses M, measure the period of ten (10) oscillations. Set the spring and mass into motion by **gently** stretching the mass 15 cm to 20 cm and then releasing it. The mass should not touch the floor or any other surface as it moves. If you pulled the mass down, then one oscillation is completed each time the mass returns to its lowest point. **Divide the total time by 10 to get the average period of one oscillation.** Record the average period for each value of M in Data Table 4.